

Spivak Chapter 13 Problems

April 8, 2026

1 Problem 1

Prove that

$$\int_0^b x^3 dx = \frac{b^4}{4}.$$

If $P = \{t_0, \dots, t_n\}$ is a partition of $[0, b]$, then

$$m_i = f(t_{i-1}) = (t_{i-1})^3 \text{ and } M_i = f(t_i) = t_i^3.$$

Subdividing this interval into n parts, we have

$$t_i = \frac{ib}{n}.$$

Then,

$$\begin{aligned} L(f, P_n) &= \sum_{i=1}^n (t_{i-1})^3 (t_i - t_{i-1}) \\ &= \sum_{i=1}^n (i-1)^3 \cdot \frac{b^4}{n^4} \\ &= \left[\sum_{i=1}^n (i-1)^3 \right] \frac{b^4}{n^4} \\ &= \left(\sum_{i=1}^{n-1} j^3 \right) \frac{b^4}{n^4}. \end{aligned}$$

Similarly,

$$\begin{aligned} U(f, P_n) &= \sum_{i=1}^n (t_i)^3 (t_i - t_{i-1}) \\ &= \sum_{i=1}^n i^3 \cdot \frac{b^4}{n^4} \end{aligned}$$

$$\begin{aligned}
&= \left[\sum_{i=1}^n i^3 \right] \frac{b^4}{n^4} \\
&= \left(\sum_{i=1}^n j^3 \right) \frac{b^4}{n^4}.
\end{aligned}$$

Because

$$\sum_{i=1}^n j^3 = \left(\frac{n(n+1)}{2} \right)^2,$$

We have

$$L(f, P_n) = \frac{b^4}{n^4} \cdot \left(\frac{n(n+1)}{2} \right)^2$$

and

$$U(f, P_n) = \frac{b^4}{n^4} \cdot \left(\frac{n(n+1)}{2} \right)^2.$$

Also

$$L(f, P_n) \leq \frac{b^4}{4} \leq U(f, P_n)$$

By choosing sufficiently large n , we can ensure that $U(f, P_n) - L(f, P_n)$ is as small as required, so

$$\int_0^b x^3 dx = \frac{b^4}{4}.$$

2 Problem 5

Evaluate without doing any computations:

1.

$$\int_{-1}^1 x^3 \sqrt{1-x^2} dx.$$

Because x^3 is odd, the integral over $[-1, 1]$ is 0.

2.

$$\int_{-1}^1 (x^5 + 3) \sqrt{1-x^2} dx.$$

First we split the integral:

$$\int_{-1}^1 x^5 \sqrt{1-x^2} dx + \int_{-1}^1 3\sqrt{1-x^2} dx.$$

Because x^5 is odd,

$$\int_{-1}^1 x^5 \sqrt{1-x^2} dx = 0.$$

$\sqrt{1-x^2}$ is the upper semicircle of radius 1, thus

$$\int_{-1}^1 3\sqrt{1-x^2} dx = \frac{3\pi}{2}.$$

3 Problem 6

Prove that

$$\int_0^x \frac{\sin(t)}{t+1} dt > 0.$$

It is clear that

$$\int_0^x \frac{\sin(t)}{t+1} dt > 0 \text{ for all } 0 \leq x \leq \frac{\pi}{2},$$

The negative of this integral is

$$\int_0^{\frac{\pi}{2}} \frac{\sin(t)}{t+1} dt$$

with $1/(t+1)$ smaller on this interval. Thus the entire integral

$$\int_{\frac{\pi}{2}}^{\pi} \frac{\sin(t)}{t+1} dt$$

is smaller than on the interval $[0, \frac{\pi}{2}]$. This is clearly also true for

$$\int_{\frac{\pi}{2}}^x \frac{\sin(t)}{t+1} dt$$

Which proves that

$$\int_0^x \frac{\sin(t)}{t+1} dt > 0.$$

4 Problem 7

Decide which of the following functions are integrable on $[0, 2]$, and calculate the integral when you can.

1.

$$f(x) = \begin{cases} x & 0 \leq x < 1, \\ x - 2 & 1 \leq x \leq 2. \end{cases}$$

$$\int_0^1 x \, dx = \frac{1}{2},$$

$$\int_1^2 x - 2 \, dx = -\frac{1}{2},$$

$$\int_0^1 x \, dx + \int_1^2 x - 2 \, dx = 0.$$

5 Problem 10

Prove, using the notation of Theorem 5, that

$$m'_i + m''_i = \inf\{f(x_1) + g(x_2) : t_{i-1} \leq x_1, x_2 \leq t_i\} \leq m_i.$$

Because $\inf A + \inf B = \inf(A + B)$,

$$m'_i + m''_i = \inf\{f(x) + g(y), t_{i-1} \leq x, y \leq t_i\}.$$

Because the infimum over all pairs (x_1, x_2) is smaller, $m'_i + m''_i \leq m_i$.

6 Problem 11

1. Which functions have the property that every lower sum equals every upper sum?

If $L(f, P) = U(f, P)$, then each $m_i = M_i$. This means that f is constant on each $[t_{i-1}, t_i]$. These intervals overlap, so f is constant on $[a, b]$.

2. Which continuous functions have the property that all lower sums are equal?

Constant functions.